CSE 2421 // Systems I

Introduction to Low-Level Programming And Computer Organization

Aaron Baxter

MoWe 8:15AM - 10:05AM

Scott Lab N050

Content

- Carmen
  - Syllabus
  - Quick Reference
  - Slides
  - Labs
  - Due Dates/Midterm/Class Schedule
- Discussion for class, labs, etc on campuswire:
  - https://campuswire.com/p/GF16F9857
  - Code: 2052
- Grades:
  - Carmen
- Lab submissions
  - CSE Unix systems and submit tool

Course Syllabus

- Review/Refer to Syllabus posted online

Grade Breakdown

- Labs: **40% (8)**
- Midterm 1: **20%**
- Midterm 2: **20%**
- Midterm 3: **20%**

- Labs will **NOT** be curved
- Exams **MIGHT** be curved at my discretion
- Overall grade will **NOT** be curved

- Extra Credit

Labs

- Labs must build on Linux systems with gcc
  - I strongly suggest you do not use IDEs
    - A lot of students write a perfect lab in Windows with Visual Studio, etc
    - Then fail the lab because they can't get it to compile in a Linux environment
- Due by 08:15:00AM on the due date
  - Friday mornings at 8:15AM
  - Late assignments automatically 20% off
  - Assignments more than 24 hours late are automatic 0s
- Submitted with Unix 'submit' system
  - Email or printed labs are not accepted
  - Timestamps are based on CSE systems

Labs - Grading

- (20%) The code and algorithm are well commented.
- (25%) The program should be appropriate to the assignment, well-structured, proper cleanup (de-allocation of resources, etc), and easy to understand without complicated and confusing flow of control.
- (50%) The results are correct, verifiable, and well-formatted. The program correctly performs as assigned.
- (5%) Documentation / Readme File

Midterms

- All midterms administered during class
- Exams are closed book, closed note, reference sheets may be provided
- Missed tests
  - You **must** inform me as soon as possible before the test if you will not be able to take it at the scheduled time
  - Missed exams without prior approval or a university approved emergency will receive a 0
  - “I overslept”, “I forgot”, “My friends were having a party”, etc. are not approved excuses

Books

- Recommended but not required
  - Computer Systems: A Programmer‘s Perspective by Bryant and O’Hallaron, 2nd ed (free online)
    - Google the above text and github
  - Pointers on C by Kenneth A. Reek
- Exams will only include material covered in class and labs

Attendance and Testing Strategy

- Attendance ***is required*** to pass the course
- Concepts, examples and material covered ***in class*** will be what is tested on
- I ***will not*** pick randomly topics from the book that we have not gone over
- Exams ***will*** cover concepts learned throughout the course
- Slides posted online are for your convenience
- This is ***not an online class***
- I ***do not*** expect to hold classes online nor intend to record lectures

Attendance for Registrar

<take attendance through Carmen>

Course Outcomes

- an ability to use the techniques, skills, and modern engineering tools necessary for practice as a CSE professional;
- an ability to design, implement, and evaluate a software or a software/hardware system, component, or process to meet desired needs within realistic constraints such as memory, runtime efficiency, as well as appropriate constraints related to economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability considerations;

How to be successful in the course

- Understand the concepts that are presented
- Ask questions
- Know the 'why' (not just how)
- Come to class and pay attention
- Ask questions about concepts you don't understand
- Google things
- Take notes on conversations and examples we do in lectures
- Do the labs and understand them

How to be Unsuccessful

- Don't come to class
- Don't pay attention
- Don't complete the labs
- Try to memorize the material
- Figure out you don't understand a concept the day before the exam
- Don't ask questions
- Don't review the material
- Don't take my advice

"Approximately one nominal hour (50 minutes) of class time and two hours for independent student work per week, per credit hour for a C average"
https://teaching.resources.osu.edu/glossary/term/credit-hours

Getting Help

- If you get confused early…You'll stay confused.
  - Please ask questions
- Asking Questions
  - Any time in class
  - Campuswire Discussion Board
  - Office Hours (after class)
- Email: baxter.95@osu.edu (administrative only)
  - Subject must contain CSE2421 (email filter)
    - I will not respond on other emails
- Anything not administrative <u>use Campuswire</u>
  - Private messages on Campuswire are ok as well
  - Campuswire is separate from all other comms for me

Getting Help - Campuswire

- Campuswire is a technological advantage and can be a useful learning tool
- Campuswire is not an excuse for attending class
- Campuswire is not the place to ask fundamental or clarification questions the night before an assignment is due. ***<u>I will reject these questions</u>***

  - ***Would you email your client/customer the night before the big proposal or project is due?***

Getting Help - Campuswire

- Campuswire will likely be answered from my phone
  - Yes or no questions can get answered quickly
- Campuswire questions will likely not be answered after 8-9pm on Thursdays
- ***Campuswire is not a replacement for asking questions during/after/before class***

Extra Credit

- Will be **HARD**
- Include extensions to topics covered in class
- Studying and doing the labs will be a better use of your time
- Probably 2 lab like programming assignments towards the end of the semester
- Up to ~10% on your final grade
- Less help available than normal labs

Extra Credit

- After the second midterm, some extra credit problems/challenges will be released, giving the opportunity to increase your grade. These extra credits will be a flat percentage added to your grade percentage after all labs and midterms have been calculated.
- However, these will only be added if your grade is at a C- or better. For example, if you have a 78% and do extra credit worth 6%, your final grade would be an 84%. If you have a 69% and do extra credit worth 12%, your final grade would be a 69%.

Feedback

- This is my sixth time teaching this course
- You'll have the opportunity to provide feedback at the end of the course (SEIs)
- But that doesn't help *you* much
- Please provide feedback *throughout* the class
  - In person (during and after class)
  - Or via email
  - Or via your friend's email, if you're afraid of the response

Waitlist

- If you anticipate dropping the course, do so immediately
  - Don’t wait until the drop deadline
- If you are not willing to working hard and learn, drop the course
  - Others are on the waitlist

Academic Misconduct

(Please don’t do it)

Academic Misconduct

- There are consequences to our actions
  - And our mistakes in particular

- The obvious:
  - I’m programming the guidance system on a missile; took a short cut; I screw up; people die.

Academic Misconduct

- The less obvious:
  - I'm programming a simple PDF parser so that people can view documents on their computer; a hostile nation-state finds a bug in my code, and quickly weaponizes it; they use the attack to break into RSA computers to steal $100M in intellectual property, which is simply a decoy; their real goals are proprietary information on SecurID token seed values, which they use to break into defense contractor systems, stealing highly classified information pertaining to national security; people die and no one knows why.

Shortcuts....

In December 2015 a glitch caused more than 3,200 US prisoners to be released early. The software calculates a prisoner's sentence depending on good/bad behaviour and was introduced in 2002.

According to reports, the problem has been ongoing for 13 years until a new IT boss was appointed and informed the governor's office. It is estimated that on average prisoners were released 49 days early.

Back in April Starbucks witnessed a register malfunction which according to Starbucks, was caused by an 'internal failure' during its routine refresh. This resulted in 60 percent of stores in the US and Canada being forced to close. The affected stores were unable to process payment transactions and at one point many stores were giving the coffee away for free.

In March 2015 F-35 Joint Strike Fighter aircrafts fell victim to a serious glitch, a software bug meant that the aircraft could not detect targets correctly. The sensors on the plane couldn't tell the difference between a singular or multiple threats.

Shortcuts....

In April 2014, a member of Google's security team found a flaw in the encryption library Open SSL, which hosts 66 percent of all websites. Although it was quickly patched by most IT firms, the sheer scale of the services affected means it is likely that there are still servers out that which remain vulnerable to attack.

WannaCry

Technical Details

The campaign uses an exploit for a recent SMB protocol vulnerability in Microsoft Windows [1, 2, 3, 7]. According to [7], the ransomware perpetrators incorporated publicly-available exploit code for the patched SMB *EternalBlue* vulnerability, CVE-2017-0145, which can be triggered by sending a specially crafted packet to a targeted SMB server. It spreads initially through vulnerable computers exposing port 445 on the Internet, and then using the same technique propagating through the internal network.

Academic Misconduct

- *This* is why…
  - It's important to *learn* the information
  - It's important to be *careful* and *know* what you are doing
  - Cheating will not be tolerated

Academic Misconduct

- I won’t hesitate to report suspected cheating
  - This is not *my* career at stake it is *yours*
  - When in doubt, *ask*
  - “I didn’t know that was cheating!” is not an acceptable excuse
- I have reported cheating multiple times

Academic Misconduct

- “By university rule, an instructor must report any suspected instance of academic misconduct to the Committee on Academic Misconduct (COAM).”
- **Read these"**
  - http://studentaffairs.osu.edu/csc/
  - http://oaa.osu.edu/coamfaqs.html
- “Cheating” goes beyond looking at someone else’s test
- A violation of course rules *is* academic misconduct

Academic Misconduct

- My rules
  - **Do not post any of your code or solutions to any course material online (Facebook, github, etc.)**
  - **Do not share your material with *anyone*, inside or outside of the course**
  - **Do not work with anyone on these assignments (friends, StackExchange, etc.)**
  - **Do not post more than 3-4 lines of code in the discussion (when in doubt, use a Private Message)**
  -

Academic Misconduct

- My rules (cont.)
  - This list is not all inclusive. **When in doubt, ask.**
  - Talking with other students is ok, **but the more you talk the closer your solutions and approaches become**

Academic Misconduct

- It is easy to cheat in CSE
- But it is really, *really* easy to catch people cheating
- Last year:
  - CSE cheating “wins” across the entire university
    - Computer Science and Engineering (8.1% of all cases)
    - Chemistry (7.5%)
    - Engineering (7.0%)
    - …
  - 531 students
  - 1226 incidents
  - 43 students from CSE
  - This is not acceptable.

Academic Misconduct

- I have a reported academic misconduct
  - Copying tests to submitting previous labs
- Co-workers have taught this class multiple times and have reported
- This makes me sad, do not sell your integrity for a few points in a class
- Academic misconduct can get you expelled from the university
- Someone stole my homework once
- It is not worth it

Academic Misconduct

- I have seen…
  - Professionals making $300k a year fired over copying a paragraph from someone else's work
  - Students destroy their entire academic careers over a homework assignment
- This cannot possibly be worth it

Academic Misconduct

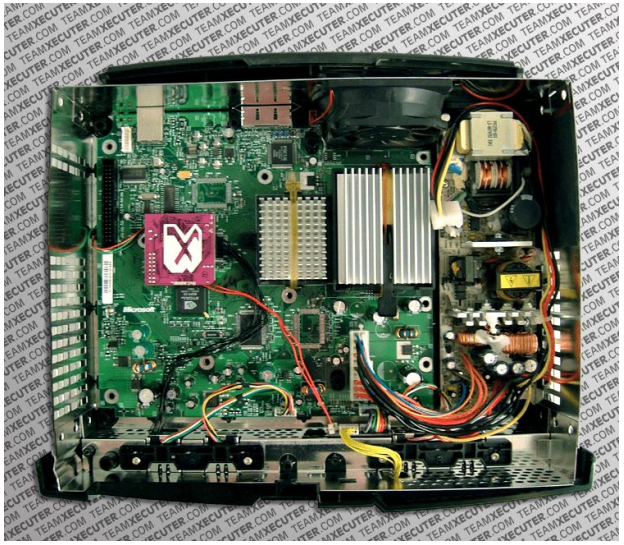
- And if you get caught…
  - Don’t ask me to let you off with a warning
  - This *is* your warning.

Myself - Personal

- From Ohio
- Wife
- 2x Son
- Dog
- House
- Hobbies:
  - Buckeyes
  - Wakeboarding/Snowboarding
  - Raspberry Pi’s / Home Automation
  - TV recording/Skipping Commercials/HDhomerun

Myself - Where/How I got started

- My Dad was always a tinkerer. Interested in early computers. Let me tinker
- Video Games
- Early "Programming" via code printed in magazines
- Played with speech processing/recognition
- Xbox (original) modifications
- Networking course in HS



Myself - Experience

- Education
  - BSECE from The Ohio State University
    - Computer Engineering/Signal Processing
  - MSECE from The Johns Hopkins University
    - Computer Architecture/Embedded systems/Hardware Design
- Experience
  - EE for Fedex Ground HQ 1.5 years
  - JHU Applied Physics Lab for 3 years in MD
  - Battelle for 10 years as “Cyber Software Engineer”
- Battelle Cyber Security R&D
  - National Security Division
  - Cyber Innovation Unit
- I do not represent Battelle - Views are my own

Myself - Work

- I work on a team of cyber security researchers
- My team specializes in many things
  - Hardware/software interaction and vulnerabilities
- My job…
  - Learn, all the time, every day
  - Write Code/Review Code
  - Manage Projects/Clients/Expectations/Budget
  - Reverse Engineer
  - Work on cool ideas and things that interest me
  - Train younger staff and interns
  - Strategize/talk with people/come up with ideas
  - Interview and hire

Myself - Philosophy

Battelle

- Right across from OSU
- World's largest non-profit R&D organization
- Manages national labs
- Contract research

Battelle

- Well known for…
  - Xerox
  - Compact Discs
  - Bar codes
  - Golf balls

This course...

This course is about…

- A programming language called C
- An assembly language called x86
- Debugging
- Computer architecture concepts
- Building foundational understanding

My friends think I code in C...

Source: Napoleon Dynamite DVD screen capture

Video Game Mods

Jailbreaking

Software Piracy

Password Cracking

```
root@sf:~/hashcat-0.46# ./hashcat-cliXOP.bin -m 1800 1800.hash rockyou.txt
Initializing hashcat v0.46 by atom with 8 threads and 32mb segment-size...

Added hashes from file 1800.hash: 14 (14 salts)

NOTE: press enter for status-screen

$6$62531178$7lty/DVyh1Kb7Xf9viQdPUmZAx.glGzw/eM3md8Da5v2.k.BHVFV7oWzj.glWS8...:123456
$6$47435678$mPiFOWkxsFDSw1q5BZo5KgLKq328F7gNYiLKarmzgBWQnX62gqEnvn.p32PQ7pC...:12345
$6$45421440$5KMHVo.EtinhoeHzbl7Cmg7K3nk18b4kLQwyN4bB6wZZ0qgDqS5XE9MQAIHzR0Z...:123456789
$6$08434354$YigIZpp3NCVxmfKO8gOTRFxieeSfLGy39xlR.T4PcOfhlvVArBzPsRq1gnQsZxN...:password
$6$14441082$2lraUIyjh6/Y7lU6f8pxL.W2q01rluNwEqX7mIjsPhe9VdQ/qpBryHjBaEMRi4m...:iloveyou
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$6$12740275$9t21hC4WgDW3yeDJ9L92LfdoyPzWEnJkAl7n.AOGpcXAO.WICN8lwcnX/HmGhiS...:daniel
$6$11072034$ODAP.JBZMdtxrg1JcjphBUPK6qmRHCyxNQgX8Kh.18940aricL6Me4/ocm.OD7o...:babygirl
$6$80867108$erLiCzZcGTNChRP3jeTqylty/6dvflXuN8/bEiR8cIStCPZjOiZ.KSA5RAKmSof...:monkey

All hashes have been recovered

Input.Mode: Dict (rockyou.txt)
Index.....: 1/5 (segment), 3560289 (words), 33550343 (bytes)
Recovered.: 14/14 hashes, 14/14 salts
Speed/sec.: - plains, 3.63k words
Progress..: 192/3560289 (0.01%)
Running...: --:--:--:--
Estimated.: 00:03:21:08

Started: Wed Jun 26 09:53:20 2013
Stopped: Wed Jun 26 09:53:20 2013
```

Charlie Miller & Chris Valasek, Car Hacking

Jay Radcliff, Insulin Death Ray

More Cyber Attacks

- A Russian hacking team known as Cold River targeted three nuclear research laboratories in the United States this past summer, according to internet records reviewed by Reuters and five cybersecurity experts.
- On December 22nd, LastPass updated a blog post with new information regarding a breach that occurred in August 2022. It now appears that customer data was implicated in the breach — specifically, a database of encrypted password vaults.
- On December 21st, Okta announced in a blog post that their source code repositories were compromised in an attack on GitHub. It is not clear how the attacker broke in.
- On November 24th, a hacker published data including email addresses and phone numbers of 5.4 million Twitter users on a hacker forum. This hacker had exploited an API vulnerability in late 2021 to scrape this data, and attempted to sell it
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- On August 17, Apple released an update to shore up iOS, iPadOS, and macOS against two security vulnerabilities: one in WebKit, which underpins Safari and other apps, and another in the kernel of the operating system itself.
- Per Apple, the Webkit vulnerability could allow malicious web pages to execute code on the device. The operating system vulnerability could allow a malicious app “to execute arbitrary code with kernel privileges”, giving it broad power over the infected device. Apple acknowledged they were “aware of a report this issue may have been actively exploited” by malicious actors, but did not go into greater detail.
- On November 24, 2014, a hacker group identifying itself as "Guardians of Peace" leaked a release of confidential data from the film studio Sony Pictures Entertainment (SPE). During the hack, the group demanded that Sony withdraw its then-upcoming film The Interview, a comedy about a plot to assassinate North Korean leader Kim Jong-un, and threatened terrorist attacks at cinemas screening the film.

(Offensive) Cyber

- Are there any *legitimate* uses to this?

(Offensive) Cyber

- But in a simpler view…
  - Need to understand offense in order to defend
    - Don’t want your hard work stolen?
    - Don’t want your software exploited?
    - Don’t want to be wirelessly killed by your medical devices?
    - Learn C.

C

- *All* of these things are possible through understanding C and assembly
  - Which is what this class is all about

C

- What if you don’t care about cyber?
  - Let’s look at other things you can do with C
    - *Or rather, things that *are generally done* with C
  - This isn’t a cyber class anyway

C

- Some things my team has worked on that were built on C…

Robot Training Simulator

- Explosive Ordnance Disposal (EOD) Robot Training Simulator (ERTS)

Robot Training Simulator

- This is the Talon robot
- It costs a few million dollars to make

Robot Training Simulator

- When it falls down a flight of stairs, it costs 10’s of thousands to fix

Robot Training Simulator

- When is it *not* a good time to learn something new?
  - When you're remotely controlling a multi-million dollar robot mounted with a .50 caliber machine gun

Robot Training Simulator

- So we took the controller…
- And virtualized the robot
- (Yep, those are xbox controllers attached to the Talon controller)

Robot Training Simulator

- This is *much* safer (and cost effective)

Robot Training Simulator

- We can now design and practice entire scenarios that would be infeasible to try in the field

Robot Training Simulator

- The graphics and physics engines for this project were written in …
  - Yep, C
  - *Real time* simulations of complex phenomena require *efficiency*
  - C is about as efficient as you can possibly get, and is the go-to language for speed

Advanced Medical Devices

Unmanned Submarines

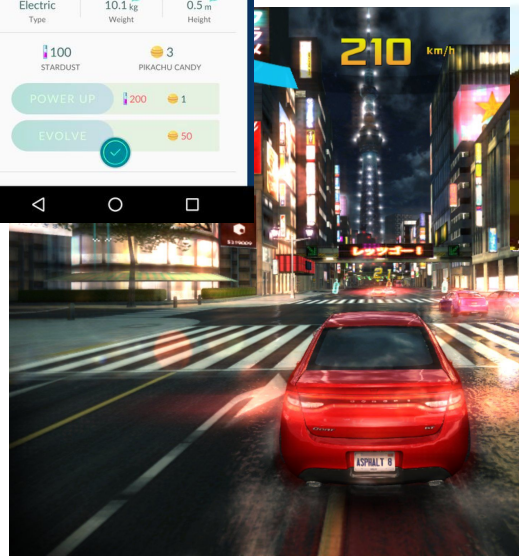
Video Games

Video Games

- Require high performance
- Physics and graphics written in C (or C++)
  - Ultra-high performance pieces written in highly optimized assembly
- GPU Shaders
  - Make games look truly incredible
  - Written in derivatives of C (HLSL, GLSL)
- How does anything look this awesome?
    - Because of C

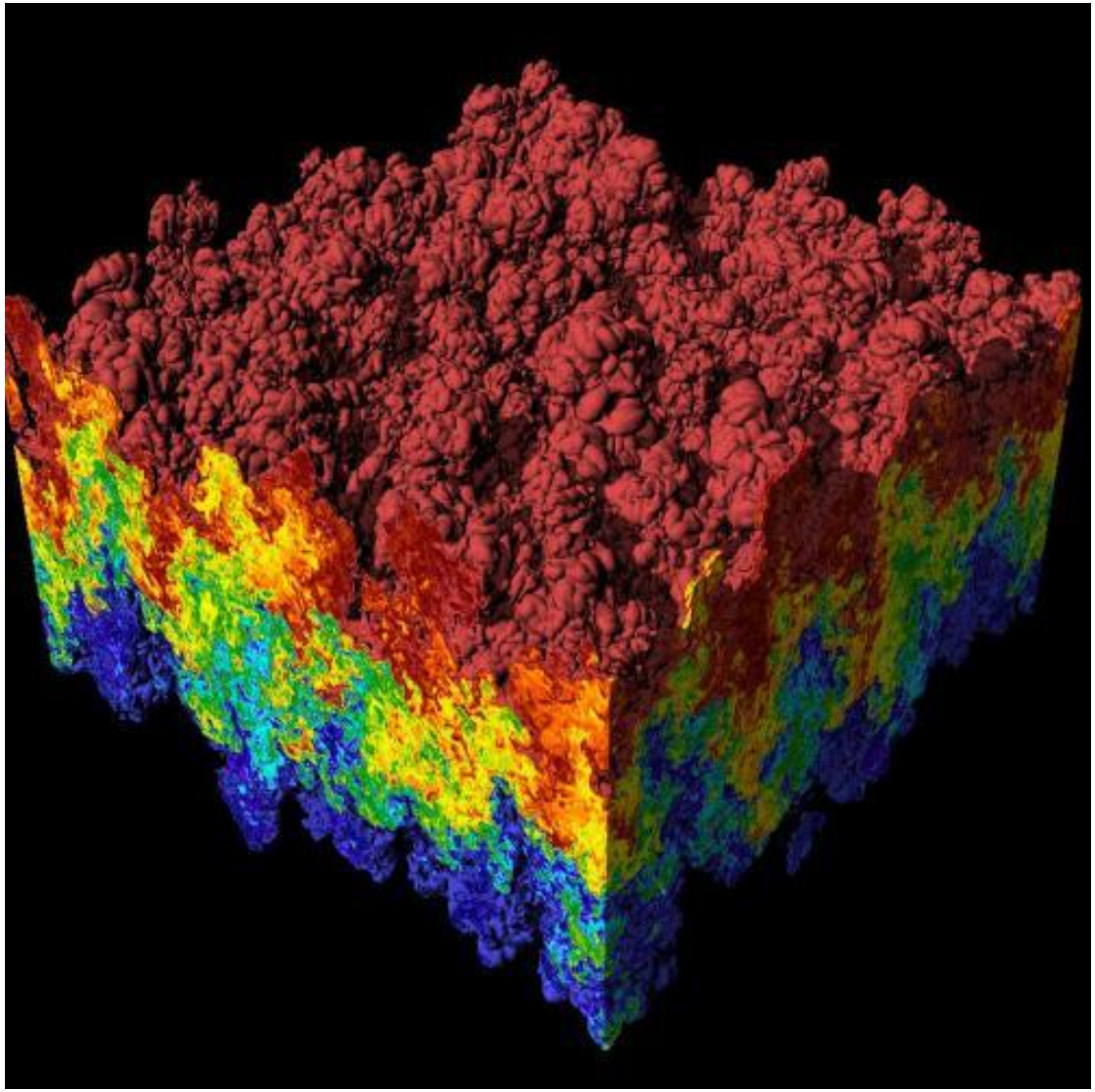
Mobile Video Games

- Augmented reality
- Reasonable graphics on constrained devices
- ~~Java can't compete~~



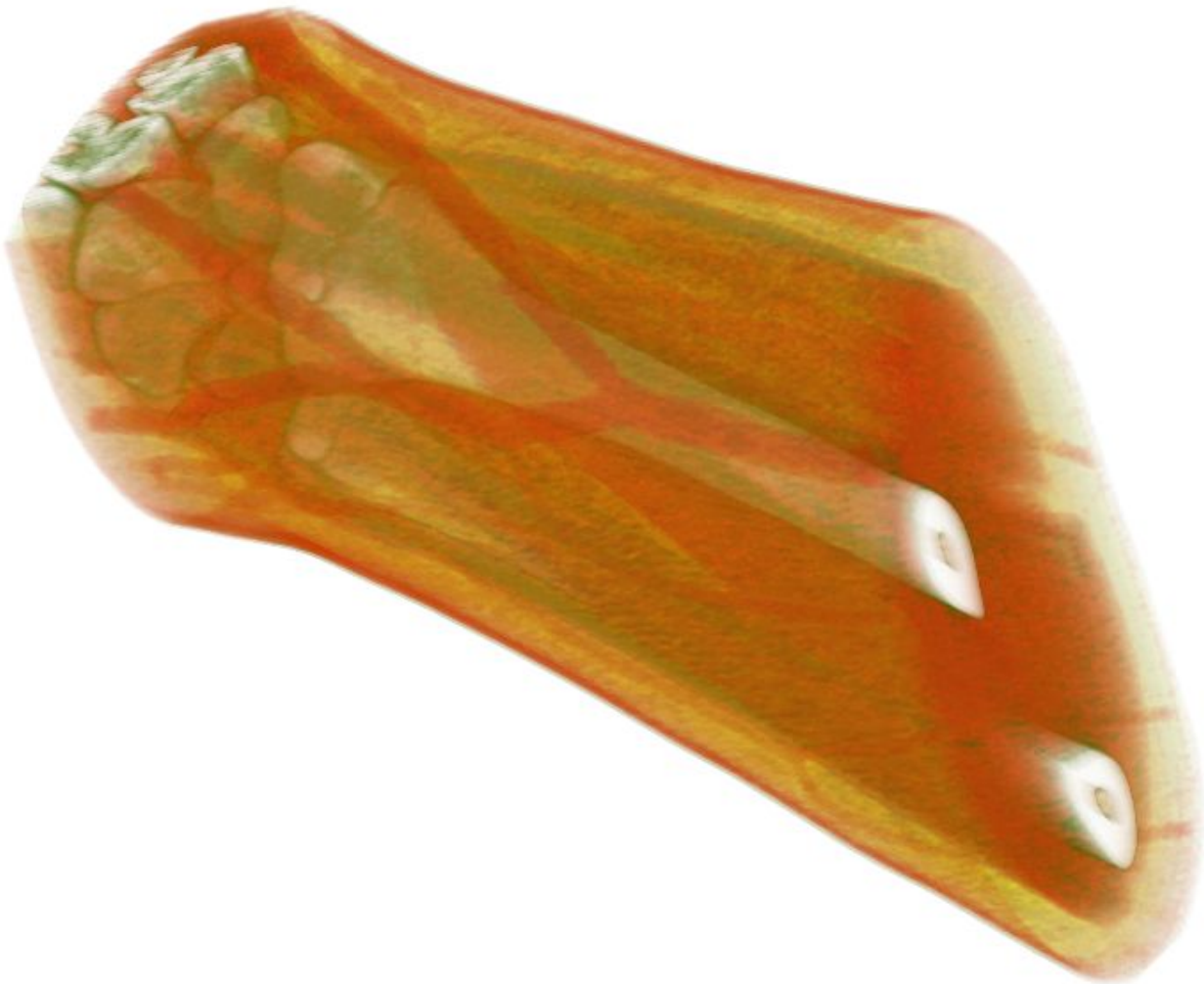
Computational Physics

- Modelling the interactions between billions of atomic and sub-atomic particles cannot be done in higher level languages



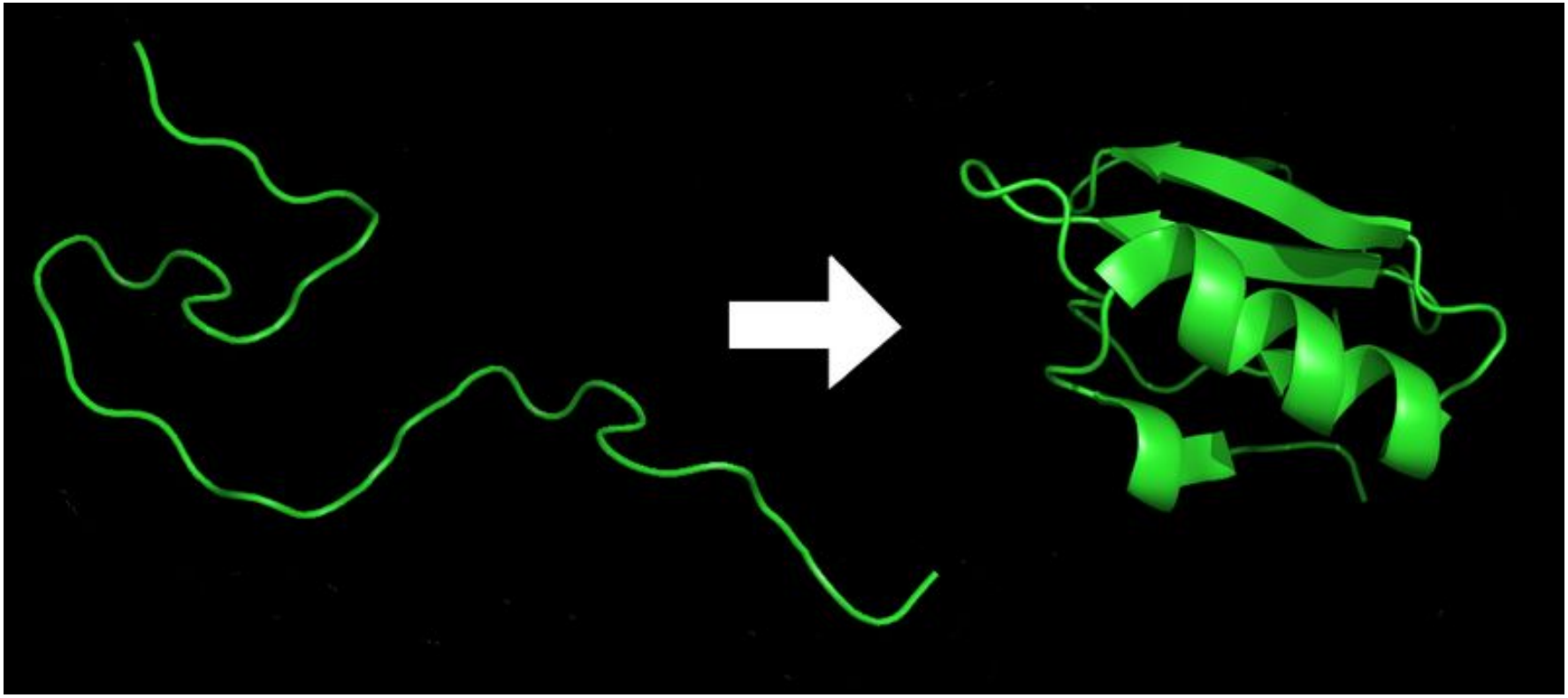
Medicine

- Volume Rendering



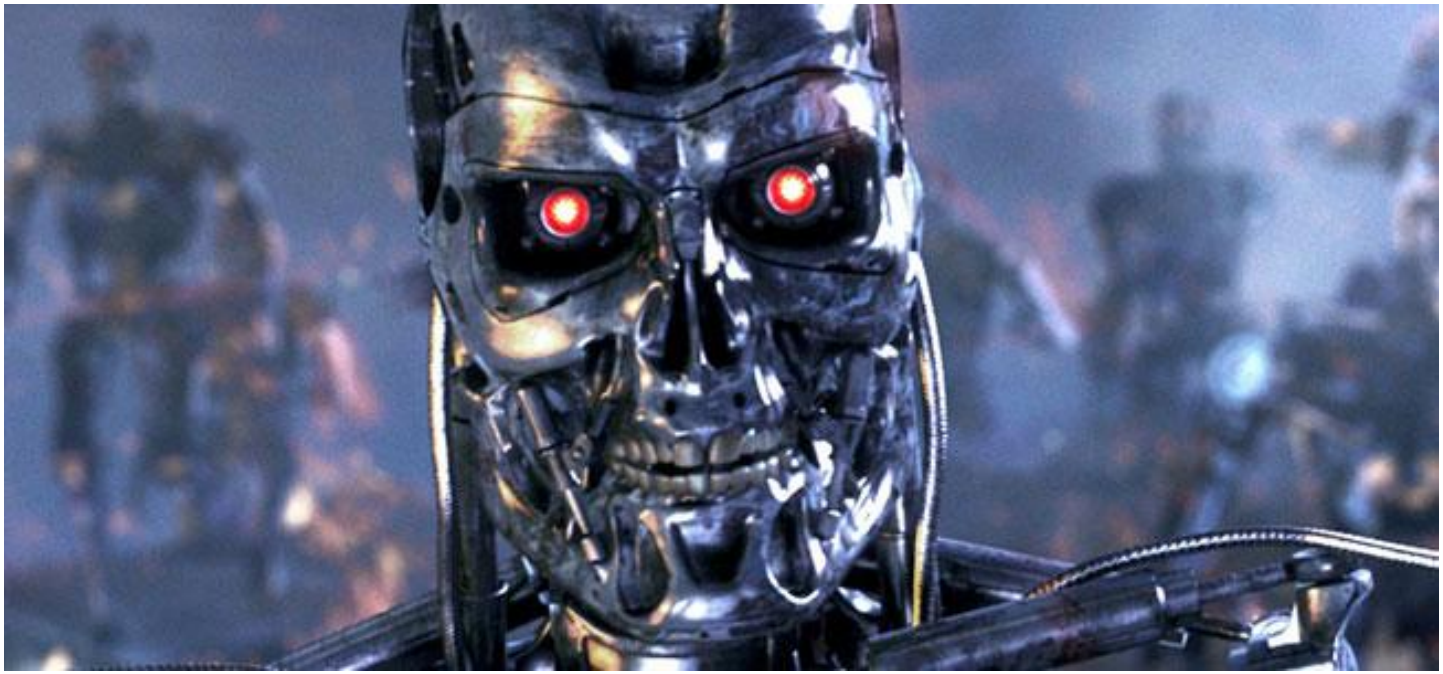
Medicine

- Protein Folding



Artificial Intelligence

- A billion neurons cannot be efficiently modeled in Java



Animation

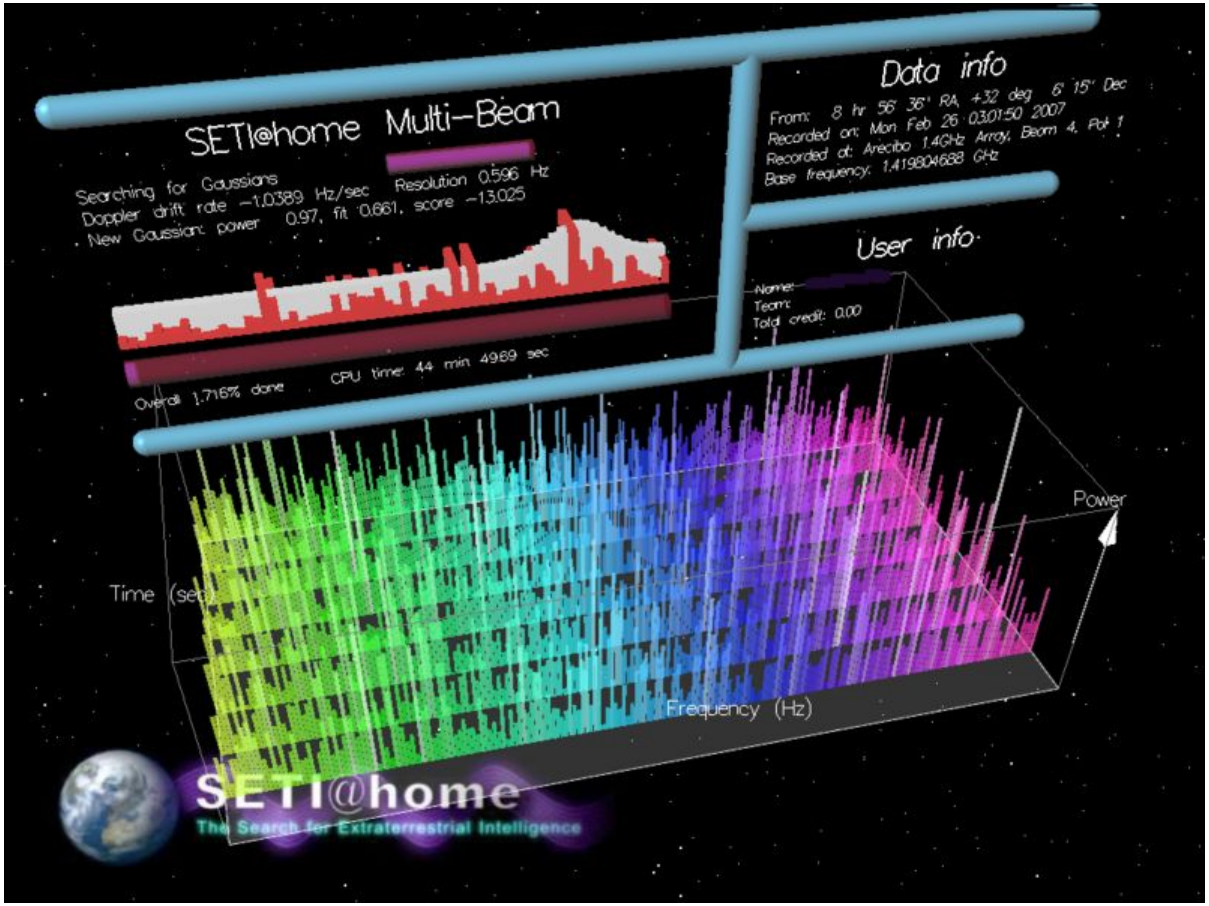
- Designing and animation can be done with high level tools written in high level languages
  - But the rendering that make it work requires extreme efficiency, and are done in C and assembly

Animation

- Final Fantasy (the movie) came out in 2001, and was really the first film of its kind – it attempted photo-realistic rendering
- It took a multi-million dollar render-farm running the most advanced and optimized C code 90 minutes to crank out a single frame
- If it were running Java, we'd still be waiting for the movie to come out

Search for Extraterrestrial Intelligence

- SETI@home
  - C, processor specific optimizations in assembly
  - Open the door for the search for extra terrestrial life



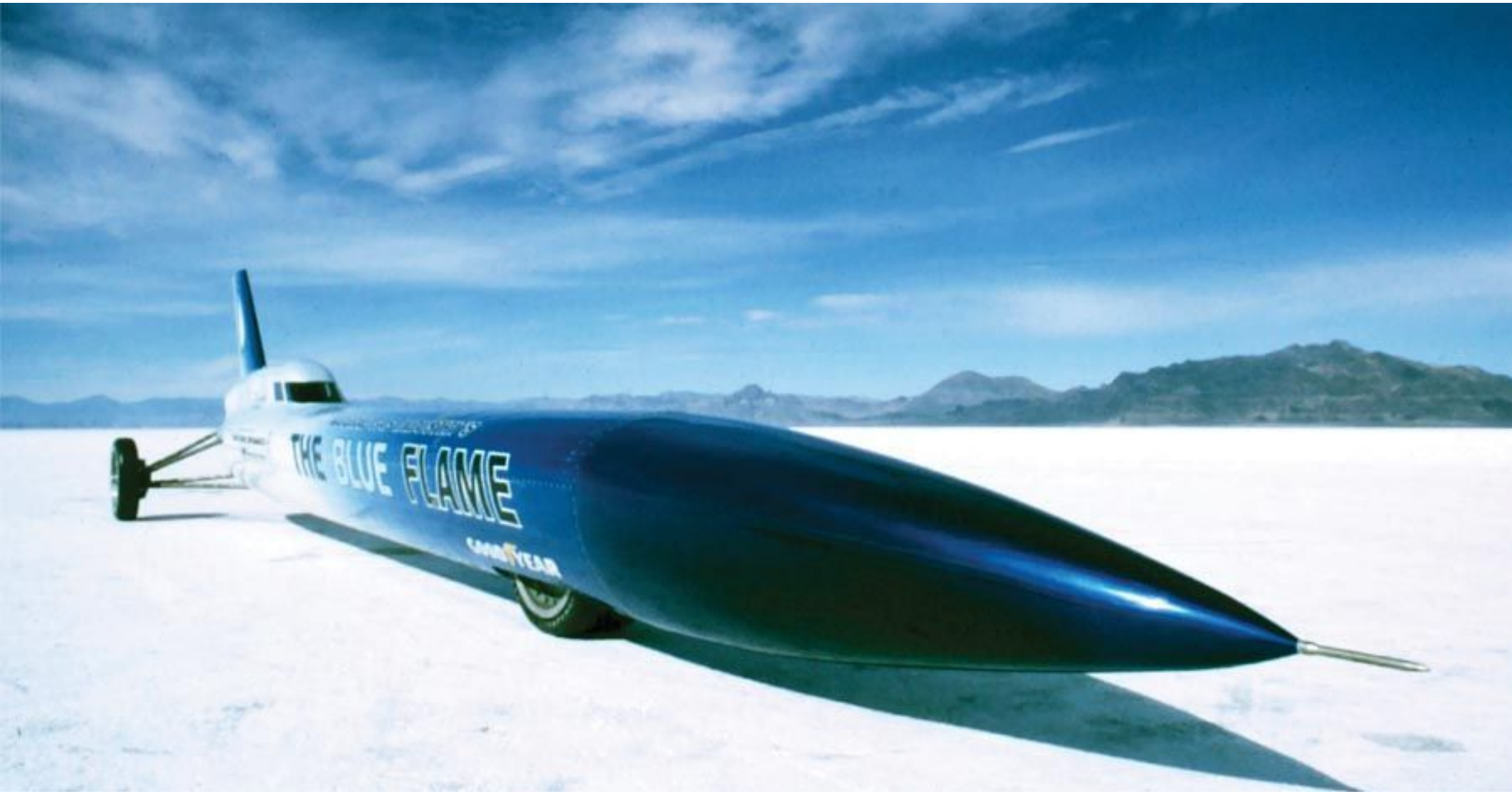
C & assembly

- Why?
  - Because they do exactly what you tell them to do
  - Compiled to Bare Metal code
  - Direct control over processing
  - No built in memory management or classes like high level languages
  - Nothing beyond what you explicitly write
  - No frills, no overhead, just raw power
  - This is what makes it challenging
  - And this is what makes it fun

Java

Java

C/assembly



C/assembly

C

- Even if you still have no interest in C/assembly…
  - The core concepts taught will be applicable to *any language and discipline*
  - Things like the stack, heap, pointers, memory management, etc, are applicable to *any* language and *any* field

Some things to take away…

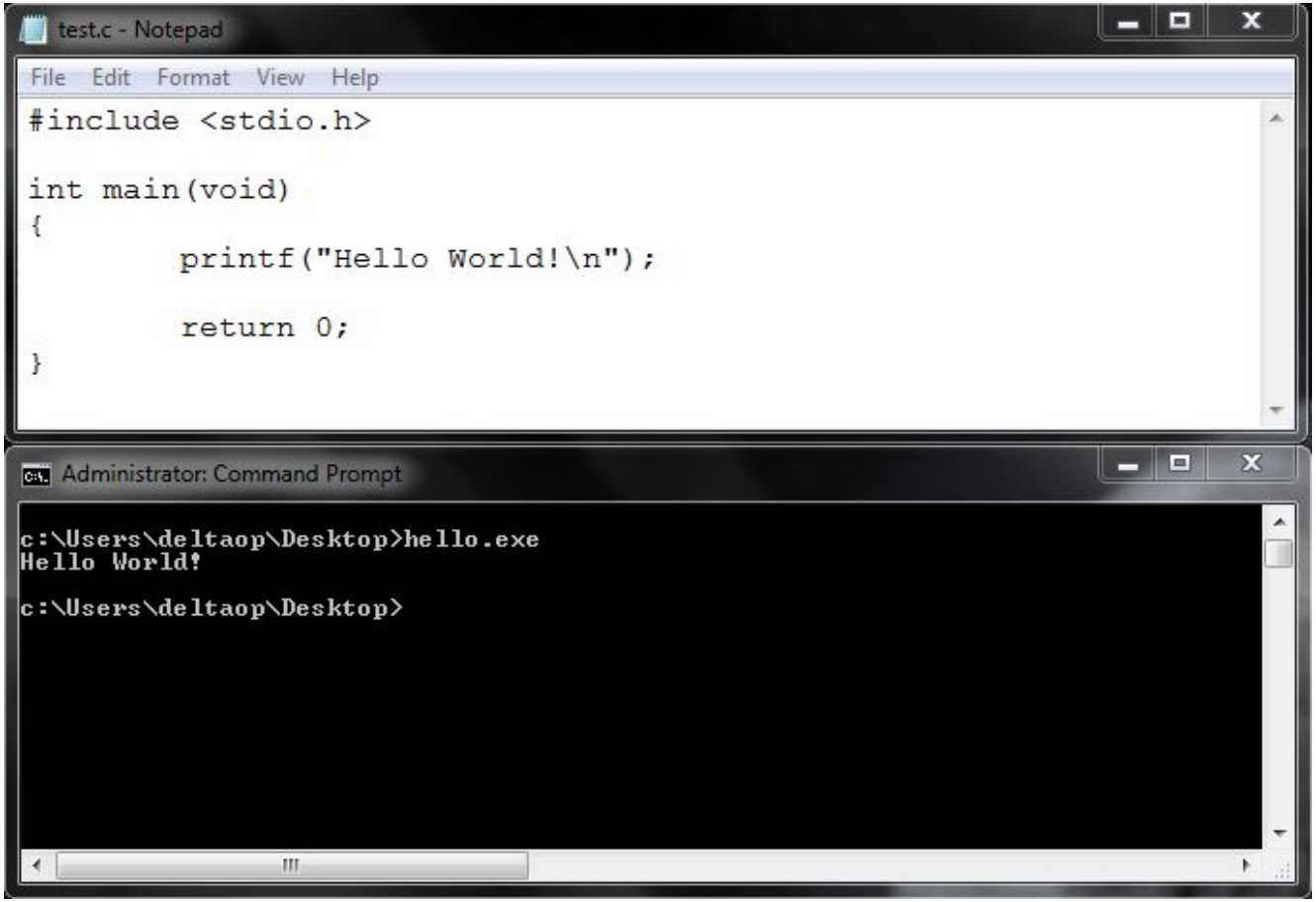
- C and assembly are used in a staggering variety of fields

Some things to take away…

- You never know what you'll fall into
  - Or what you might enjoy

Some things to take away…

- So while it's not *always* the most exciting language…



Some things to take away…

- It *can* be.

Some things to take away…

- So throughout this course…
  - Stay focused on the end applications
  - I'll try not to waste your time
  - But if you're ever in doubt as to why something matters
    - Just ask!

Lab0

Lab0 due Friday Morning!
Lab0 due Friday Morning!
Lab0 due Friday Morning!
Lab0 due Friday Morning!

Lab0

Get signed into your CSE Linux account
Get acquainted with Linux
Get acquainted with Text editors
Get acquainted with ssh/scp
Get acquainted with the terminal
Get acquainted with the submit tool
Create your hash for the semester